

Canonical Flow and Bottleneck Condensation in Finite Gordon–Newell Networks

Route-A source-local certificate HCS-C285

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Abstract

For every finite irreducible routing matrix, without assuming reversibility, we give a self-contained closure of the single-class Gordon–Newell network. The canonical partition function is the complete homogeneous polynomial $Z_N = h_N(w)$; its derivatives give every joint occupancy moment, while Z_{N-1}/Z_N gives every station throughput and directed service-event flow. The stationary reverse routing is obtained exactly and yields a sharp reversibility criterion. Gordon and Newell own the classical product-form and bottleneck lineage; no literature originality, arithmetic determinant, or quantization is claimed.

1 Frozen owner and classical lineage

Fix $m \geq 1$, $N \in \mathbb{Z}_{\geq 0}$, positive service rates μ_i , and an irreducible row-stochastic matrix $P = (p_{ij})$. Zero entries and self-routes are allowed; P need not satisfy detailed balance. On

$$\mathcal{S}_N = \{n \in \mathbb{Z}_{\geq 0}^m : \sum_i n_i = N\},$$

an occupied station i completes service at total rate μ_i and sends the customer to j with probability p_{ij} . Thus

$$Lf(n) = \sum_{i: n_i > 0} \mu_i \sum_j p_{ij} [f(n - e_i + e_j) - f(n)]. \quad (1)$$

When $i = j$, a service event occurs but the state does not change. This distinction matters for physical edge flows below.

Let e be the unique positive traffic vector with $e = eP$ and $\sum_i e_i = 1$, and put $w_i = e_i/\mu_i$. Gordon and Newell’s paper is the classical equilibrium and bottleneck owner [1]. Kelly supplies the standard reversal framework [2], and Kelly–Yudovina give a modern queueing-network account [3]. Those sources are cited for classical ownership and standard context; no literature originality is claimed. The proofs below are included to freeze the exact conventions and boundaries used by the certificate.

2 Canonical law and all occupancy derivatives

Define

$$Z_N = h_N(w) = [z^N] \prod_{i=1}^m (1 - w_i z)^{-1} = \sum_{n \in \mathcal{S}_N} \prod_i w_i^{n_i}, \quad Z_0 = 1. \quad (2)$$

Theorem 1 (finite Gordon–Newell closure). *The unique stationary probability is*

$$\pi_N(n) = Z_N^{-1} \prod_i w_i^{n_i}. \quad (3)$$

For $D_i = w_i \partial_{w_i}$,

$$\mathbb{E} n_i = D_i \log Z_N, \quad \text{Cov}(n_i, n_j) = D_i D_j \log Z_N. \quad (4)$$

For every multi-index $\alpha \in \mathbb{Z}_{\geq 0}^m$,

$$\mathbb{E} \prod_i (n_i)^{\alpha_i} = \frac{w^\alpha \partial_w^\alpha Z_N}{Z_N}. \quad (5)$$

Thus (5), followed by Stirling conversion, gives all joint ordinary moments.

Proof. For a fixed n , the outgoing nonself rate is $\sum_{j:n_j>0} \mu_j(1-p_{jj})$. An incoming jump with destination j starts from $n - e_j + e_i$ and has weight ratio w_i/w_j . Its total stationary mass for this j equals

$$\pi_N(n) \sum_{i \neq j} \frac{w_i}{w_j} \mu_i p_{ij} = \pi_N(n) \frac{\mu_j}{e_j} \sum_{i \neq j} e_i p_{ij} = \pi_N(n) \mu_j (1 - p_{jj}),$$

where $e = eP$ was used in the last equality. Summing over occupied j proves global balance without reversibility. Strong connectivity makes the composition chain irreducible when $N > 0$; $N = 0$ is a singleton. Differentiating the finite sum (2) proves (4)–(5). $\square$

Equation (2) also gives two independent recurrences useful for exact reconstruction. If $p_k = \sum_i w_i^k$, then

$$Nh_N = \sum_{k=1}^N p_k h_{N-k}, \quad h_0 = 1, \quad (6)$$

while multiplying the m truncated geometric series gives a convolution recurrence. Neither recurrence divides by $w_i - w_j$, so equal weights are regular.

3 Throughput, directed currents, and time reversal

Put $R_N = Z_{N-1}/Z_N$ for $N \geq 1$ and $R_0 = 0$. Decrementing coordinate i in (2) gives

$$\mathbb{P}(n_i > 0) = w_i R_N, \quad T_i = \mu_i \mathbb{P}(n_i > 0) = e_i R_N, \quad J_{ij} = e_i p_{ij} R_N. \quad (7)$$

Here J_{ij} counts service events, including $i = j$. Traffic balance gives $\sum_j J_{ij} = \sum_j J_{ji} = T_i$; hence $C_{ij} = J_{ij} - J_{ji}$ is an antisymmetric divergence-free current.

Define

$$p_{ij}^* = \frac{e_j p_{ji}}{e_i}. \quad (8)$$

Theorem 2 (exact reversal and iff criterion). *P^* is irreducible and row-stochastic, has traffic vector e , and is the routing matrix of the stationary time-reversed occupancy process with the same service rates. Reversal is involutive. For $N \geq 1$, the state process is reversible if and only if*

$$e_i p_{ij} = e_j p_{ji} \quad \text{for every } i, j, \quad (9)$$

equivalently $P^* = P$. At $N = 0$ the singleton chain is trivially reversible for every routing matrix and therefore cannot identify (9).

Proof. If $n' = n - e_i + e_j$, stationary rate reversal and (3) give

$$q^{\text{rev}}(n', n) = \frac{\pi_N(n) \mu_i p_{ij}}{\pi_N(n')} = \mu_j \frac{e_i p_{ij}}{e_j} = \mu_j p_{ji}^*.$$

The traffic equations imply the row sums and stationary vector of P^* ; the same formula applied twice returns P . Positive population supplies a state transition along every routing edge, so detailed balance of the state chain is equivalent to (9). Reversed event flow is $J_{ij}^* = J_{ji}$. $\square$

References

- [1] William J. Gordon and Gordon F. Newell, *Closed Queuing Systems with Exponential Servers*, Operations Research 15(2) (1967), 254–265, [doi:10.1287/opre.15.2.254](https://doi.org/10.1287/opre.15.2.254).
- [2] F. P. Kelly, *Reversibility and Stochastic Networks*, Wiley, Chichester, 1979; Cambridge University Press reissue, 2011, ISBN 978-1-107-40115-0, [authorized author-hosted edition](https://doi.org/10.1017/CBO9781139565363).
- [3] Frank Kelly and Elena Yudovina, *Stochastic Networks*, Cambridge University Press, 2014, [doi:10.1017/CBO9781139565363](https://doi.org/10.1017/CBO9781139565363).